DATE:

Ex. No. 1 **CHARACTERISTICS OF SCR**

AIM :

To find the latching and holding current from the VI characteristics of Silicon Controlled Rectifier.

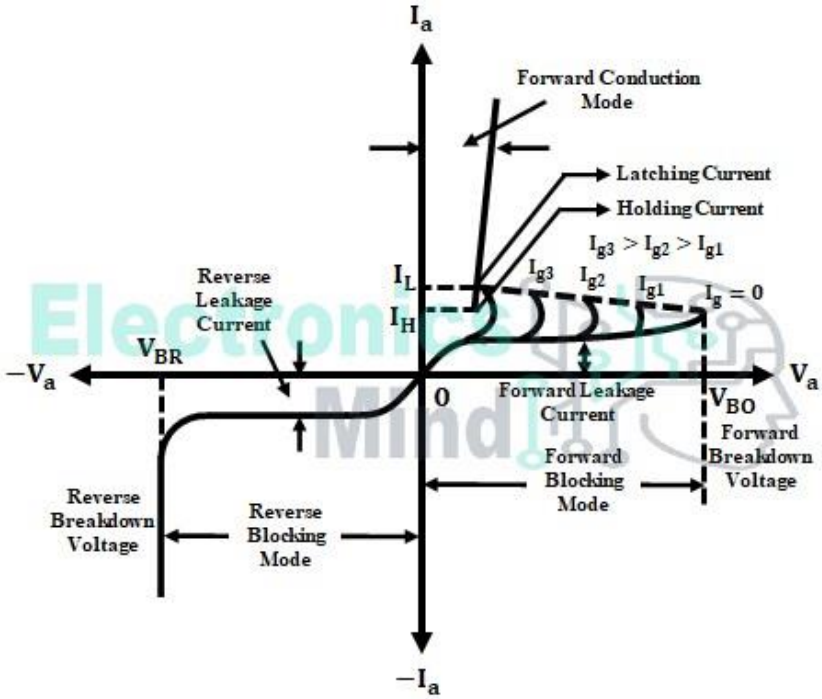
APPARATUS REQUIRED:

S.No.	APPARATUS	RANGE	TYPE	QUANTITY
1	SCR	TYN 604		1
2	Regulated Power Supply	(0-30) V		1
3	Voltmeter	(0-20) V	MC	1
4	Ammeter	(0-100)mA	MC	1
5	Ammeter	(0-100)μA	MC	1
6	Resistor	1 kΩ		2
8	Connecting wires			Few

THEORY:

An SCR is a 4-layer, 3-junction, 3-terminal device. When anode is positive with respect to cathode, the curve between V_{AK} and I_A is called the forward characteristics. During forward bias condition, the junction J_2 is reverse biased and when across J_2 above break over voltage (V_{BO}), J_2 breaks down and heavy current will flow in the device. Hence a load resistance is always connected in series with the SCR to limit the anode current to safe value. Latching current is the minimum anode current required to turn ON SCR without gate current? Holding current is the maximum anode current at which SCR turns OFF from ON condition, with gate open.

MODEL GRAPH



TABULATION:

S.No	V_{AK} (V)	I_A (mA)

PROCEDURE:

To Plot V.I Characteristics:

1. Make the connections as per the circuit diagram.
2. Switch ON the regulated power supply. Apply some constant voltage say 30V by varying V_{AK} source.
3. Gradually increase the gate current by varying V_{GK} source till the SCR becomes ON. Note down the corresponding value of I_G from the milliammeter. Then decrease V_{AK} and V_{GK} to minimum.
4. Set gate current equal to noted value in step 3 by varying V_{GK} source.
5. Gradually increase V_{AK} in steps of 2V and for each step note down the value of V_{AK} and I_A , and then reduce V_{AK} to minimum.
6. Set gate current to some other value (preferably higher than that of the value set in step 3)
7. Repeat step 5.
8. Plot a graph of V_{AK} versus I_A for different values of I_G.

To Find Latching current (I_L):

1. Keep proper V_{AK} to trigger SCR by gate current. Then trigger SCR by applying gate current.
2. Gradually decrease V_{AK} in steps and at each step switch-off the gate supply (i.e. V_{GK} source) an observe that, whether device remains in the ON state or not.
3. Repeat step 2 (by trial and error method) till the SCR jumps to blocking state, and then note down the minimum value of I_A which keeps device in the on state as Latching current.

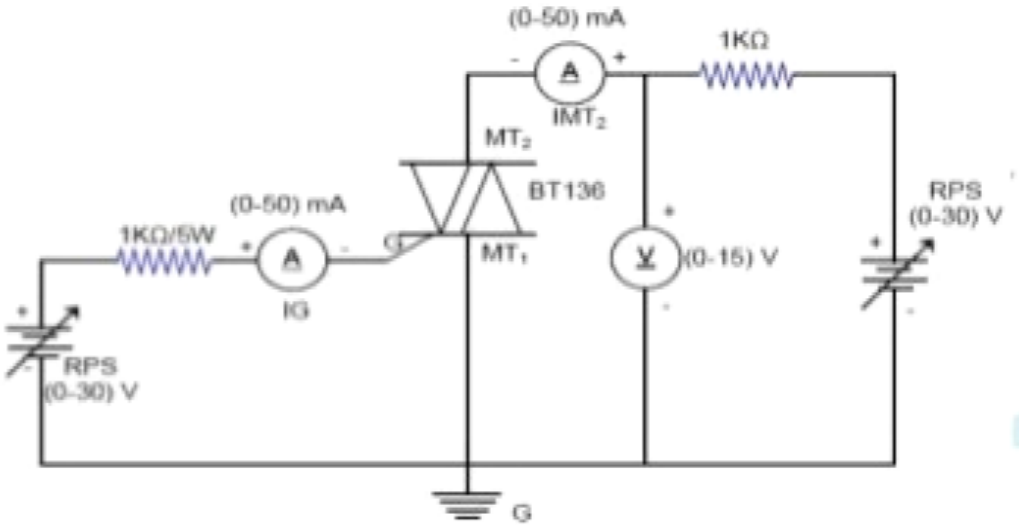
To Find Holding current (I_H):

1. Keep proper V_{AK} to trigger SCR by gate current. Then trigger SCR by applying gate current.
2. Switch-Off V_{GK} source permanently. Now gradually decrease V_{AK} and note down the minimum value of I_A below which, the device suddenly falls from ON-state to OFF- state as Holding current.

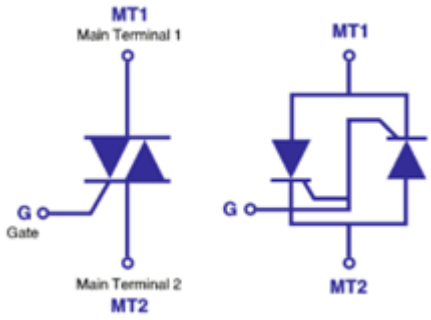
RESULT:

The characteristics of SCR is done and latching current and holding current is found from the characteristics

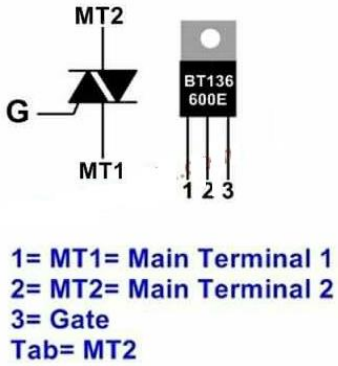
CIRCUIT DIAGRAM



TRAIC SYMBOL



TRAIC CONFIGURATION



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Ex. No. 2 **CHARACTERISTICS OF TRIAC**

AIM:

To obtain the following static characteristics of a given TRIAC

(a) Forward conduction characteristics (Va Vs Ia)

(b) Reverse conduction characteristics (Va Vs Ia in reverse direction)

A**PPARATUS REQUIRED**:

S.No.	APPARATUS	RANGE	TYPE	QUANTITY
1	Triac	-	-	1
2	Regulated Power Supply	(0-30) V		1
3	Voltmeter	(0-20) V	MC	1
4	Ammeter	(0-100)mA	MC	1
5	Ammeter	(0-100)μA	MC	1
6	Resistor	470Ω,560Ω		1
7	Connecting wires			Few

PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Fix the gate current by varying RPS at gate for forward direction.
3. With the help of RPS at MT_1 to MT_2 side note down the values of Ia for various values of Va.
4. The above steps are repeated for different values of $I_{G.}$
5. Draw the graph between Va and Ia
6. Repeat the same with terminals reversed to obtain the reverse characteristics of TRIAC
7. For reverse direction change MT_1 and MT_2 terminals or RPS and repeat from STEP 2 to 6 and note down the readings for reverse direction